High-Order Winslow smoothing/untangling

Andrew Shi

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Codes live here.

0 The 2016 formulation (brief recap)

The Winslow equations in physical coordinates are

$$g^{ij}\,\partial_i\partial_j x_k = 0, \qquad k = 1, \ldots, n, \tag{1}$$

where g^{ij} are the contravariant metric components of the (nonlinear) mapping from the computational domain C to the physical domain D. The nonconservative form does not discretize cleanly in a CG framework, so the 2016 paper introduces auxiliary variables $\alpha_j = -\partial_i(g^{ij})$ and recasts the system in split form:

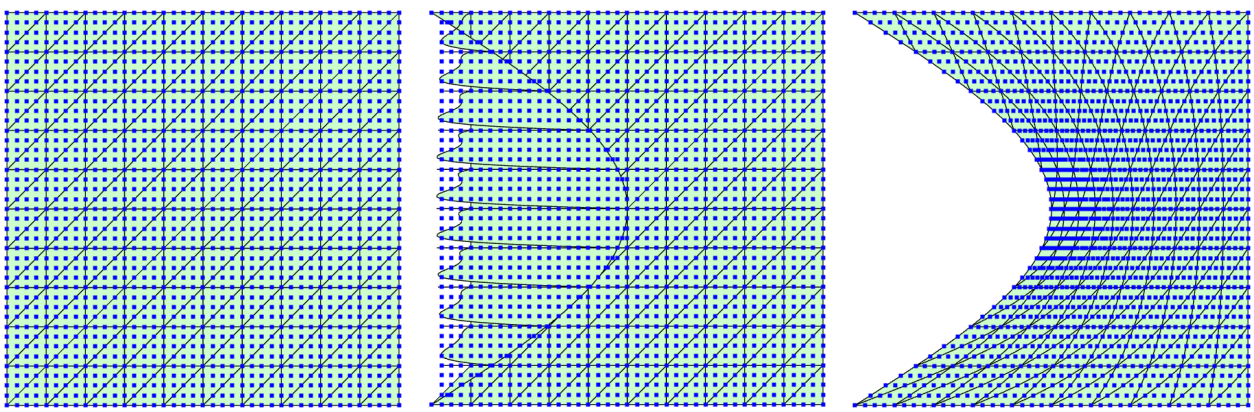
$$\partial_i(g^{ij}) + \alpha_j = 0, \tag{2}$$

$$\partial_i(g^{ij}\,\partial_j x_k) + \alpha_j\,\partial_j x_k = 0, \tag{3}$$

with Dirichlet data $\boldsymbol{x} = \boldsymbol{x}_{\text{bnd}}$ on ∂C. The point of this split form is to 1) mimic the what finite difference Winslow people are doing and 2) enable Picard iteration, which is the "magic" here.

0.1 Numerical Results/Tests

The code `w2.m` does the following in Figure 1.

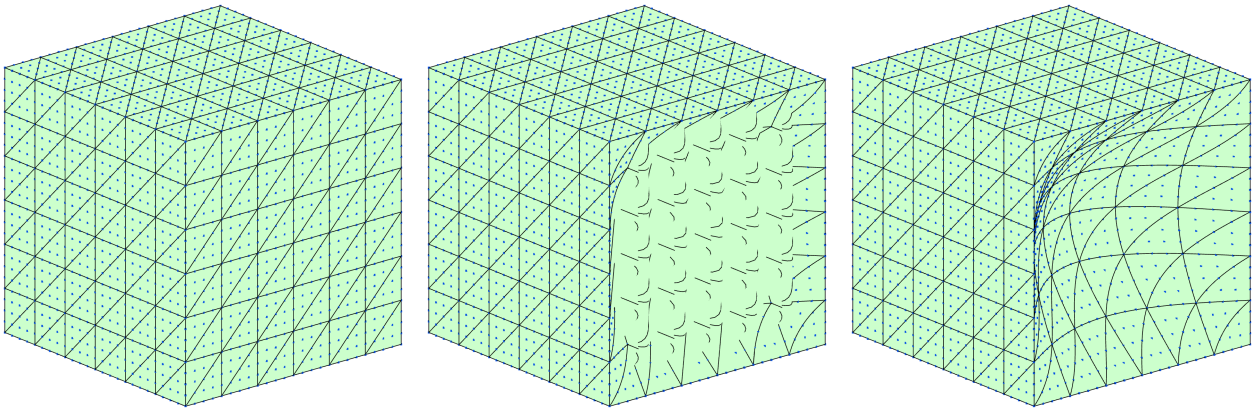


(a) Reference mesh. (b) Initial configuration. (c) After Winslow

Figure 1: (a) Structured 11×11 reference mesh at $p = 4$. (b) Initial configuration: interior nodes unchanged, boundary on side 4 displaced by $0.5\sin(\pi y)$, producing heavy tangling. (c) After Winslow iterations, interior nodes have redistributed to absorb the boundary deformation; all elements valid.

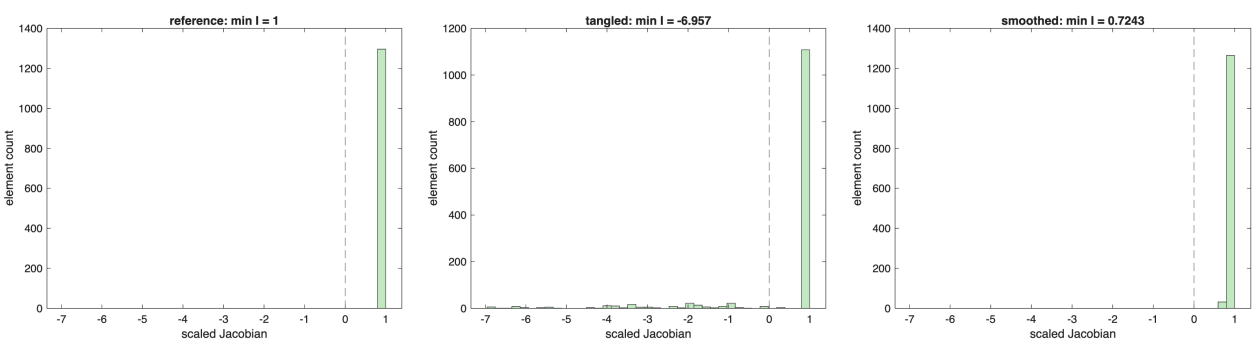
It is straightforward to ask Claude Code to write the 3D analogue `w2_3d.m` in in Figure 2. The histograms of the scaled Jacobians are given in Figure 3.

These notes record the starting state of the W-modified Winslow smoothing project and fix notation for what comes next. The plan is to extend Fortunato–Persson 2016 [**?**] by introducing a per-element target-shape matrix $\boldsymbol{W}$ into the Winslow system so that the physical mesh shape is governed by $\boldsymbol{W}$ rather than inherited from the reference mesh. Before introducing $\boldsymbol{W}$, this note replicates the 2016 paradigm on a stress test so we have a clean baseline.



(a) Reference mesh. (b) Initial configuration. (c) After Winslow

Figure 2: (a) Structured $7\times7\times7$ tetrahedral reference mesh of the unit cube at $p = 4$. (b) Initial configuration: interior nodes unchanged, boundary on side 1 ($x = 0$) displaced by $0.4\sin(\pi y)\sin(\pi z)$, producing heavy tangling. (c) After Winslow iterations, interior nodes have redistributed to absorb the boundary deformation; all elements valid.



(a) Reference (b) Tangled (c) Smoothed

Figure 3: Per-element scaled Jacobian $I = \min_\xi J(\xi)/\max_\xi J(\xi)$ (Fortunato–Persson 2016, Sec. 4) for each configuration. (a) Reference: all elements straight-sided simplex, $I = 1$. (b) Tangled: long negative tail with $\min I = -6.96$; boundary-adjacent elements inverted. (c) Smoothed: all $I > 0$, most near 1, with $\min I \approx 0.70$ on elements bordering the indentation.

1 Test setup

A structured 11×11 reference mesh of the unit square is built at polynomial degree $p = 4$. The nodes on one boundary edge (side 4) are displaced by $x \mapsto x + 0.5\sin(\pi y)$, while all other boundary nodes and all interior nodes stay at their reference positions. With grid spacing $\Delta = 0.1$, the boundary perturbation amplitude of 0.5 cuts through roughly five element layers, producing heavy tangling near the perturbed boundary. The Winslow–Picard solver from the 2016 paper (`elliptic_smoothing.m`) is then applied, with the reference mesh as the computational domain and the perturbed node positions supplied as Dirichlet data on the physical coordinates.

The driver script `w2.m` is reproduced below.

```
porder = 4;
n = 10;
msh = mshsquare(n+1, n+1);
msh = nodealloc(msh, porder);

% Output directory for figures
figdir = 'figures';
if ~exist(figdir, 'dir'), mkdir(figdir); end

% --- Figure 1: reference mesh -------------------------------------------
figure(1); clf;
dgmeshplot_curved(msh, 4, 1, 0);
set(gcf, 'Position', [114 1 560 420]);
exportgraphics(gcf, fullfile(figdir, 'mesh_reference.png'), 'Resolution', 200);

% --- Build the perturbation: displace nodes on boundary 4 ---------------
[~, edg] = getbndnodes(msh, dginit(msh), 4);
x = msh.p1(:,1,:);
y = msh.p1(:,2,:);
newx = x;
newy = y;
newx(edg) = x(edg) + 0.5*sin(pi*y(edg));
curvep1 = cat(2, newx, newy);

msh_tangled = msh;
msh_tangled.p1 = curvep1;

% --- Figure 2: tangled initial configuration ----------------------------
figure(2); clf;
dgmeshplot_curved(msh_tangled, 4, 1, 0);
set(gcf, 'Position', [114 1 560 420]);
exportgraphics(gcf, fullfile(figdir, 'mesh_tangled.png'), 'Resolution', 200);

% --- Run Winslow-Picard -------------------------------------------------
doplot = false;
msh1 = elliptic_smoothing(msh, curvep1, doplot);

% --- Figure 3: final smoothed mesh --------------------------------------
figure(3); clf;
dgmeshplot_curved(msh1, 4, 1, 0);
set(gcf, 'Position', [114 1 560 420]);
exportgraphics(gcf, fullfile(figdir, 'mesh_smoothed.png'), 'Resolution', 200);
```

2 Results

3 Observations

The 2016 method untangles this configuration within its expected regime: the tangling is boundary-induced, spatially localized, and topologically compatible with a valid target. This is exactly the regime Fortunato–Persson 2016 treats. The contribution of the present project is not to extend what the 2016 method can untangle (it cannot untangle genuinely tangled references); the contribution is to change what the resulting physical mesh *looks like*. Via $\boldsymbol{W}$-modification, the target element shape will be decoupled from the reference element shape, so a skewed reference no longer forces skewed physical elements.

4 Next steps

- Derive the $\boldsymbol{W}$-modified system. With $\boldsymbol{J}_{\text{total}} = \boldsymbol{J}\boldsymbol{W}^{-1}$, the modified contravariant metric is $\tilde{g}^{ij} = W_{ik}\, g^{kl}\, W_{jl}$, and the split-form equations retain their structure with g^{ij} and α_j replaced by their tilded counterparts.
- Implement the $\boldsymbol{W}$-modified α-projection and Picard solve in 2D. Verify that Picard iteration counts are comparable to the unmodified case on a smooth test problem.
- Repeat the stress test in this note with a *skewed* reference mesh. Compare post-convergence element quality (scaled Jacobian histogram) against unmodified Winslow on the same reference. This is the minimum test that would establish $\boldsymbol{W}$-modification as a real capability extension over 2016.